

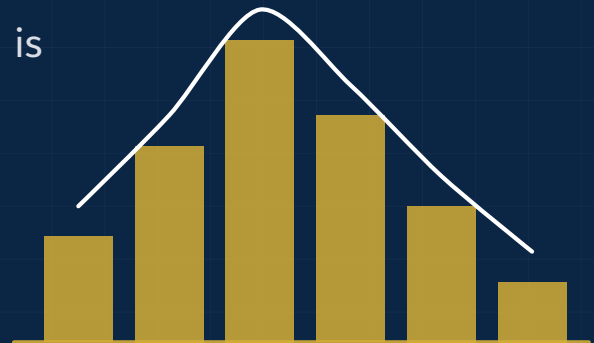


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CHAPTER 6 • STATISTICS & PROBABILITY • TOPIC 2

Histograms

In a histogram the AREA of a bar is the frequency. Height = frequency density = frequency $\div$ class width. Every question is one of those two lines.



3

VARIANTS

26

PAST-PAPER PARTS

40

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Frequency density & drawing

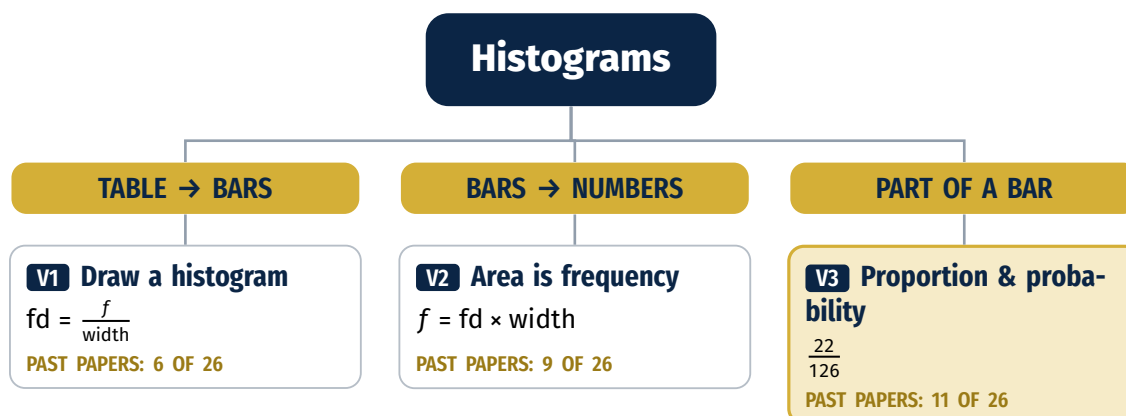
V2 Area is frequency

V3 Part of a bar: proportion & probability

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Histograms at a glance



Your notes


THE KEY FACT

AREA = FREQUENCY

Not the height — wide bars hold more.

★ How often does each variant come up?

20 past-paper questions (Jan 2020 → Nov 2025) hold 26 parts; 40 practice parts — grouped by type.

	PAST PAPERS (OUT OF 26)	PRACTICE PARTS
V1 Frequency density, drawing	6	15
V2 Area is frequency	9	19
V3 Part of a bar: proportion	11	6

WHAT THIS TELLS YOU On almost every paper (Q14–22, 3–5 marks). Often the scale is missing: one bar with a known frequency tells you what one square is worth. Then it is all area.

BEFORE YOU START

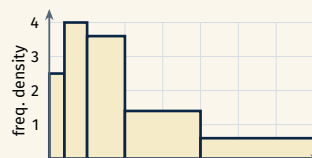
The big idea

**Area, not height, is the frequency.**

Classes can have different widths. A wide class would look too big if the height were the frequency, so the height is the **frequency density**:

$$fd = \frac{\text{frequency}}{\text{class width}}$$

Then **area** = width × fd = frequency.


The two formulas
Table → bar

$$fd = \frac{f}{\text{width}}$$

The height to draw.

Bar → table

$$f = fd \times \text{width}$$

The area of the bar.

No scale on the axis?
Known bar

20 trains in 10–15

Its area in squares.

One square

$\frac{20}{\text{squares}}$

Worth this many.

Other bars

count squares

× the value of one square.

**CALCULATOR FIRST, THEN CHECK**

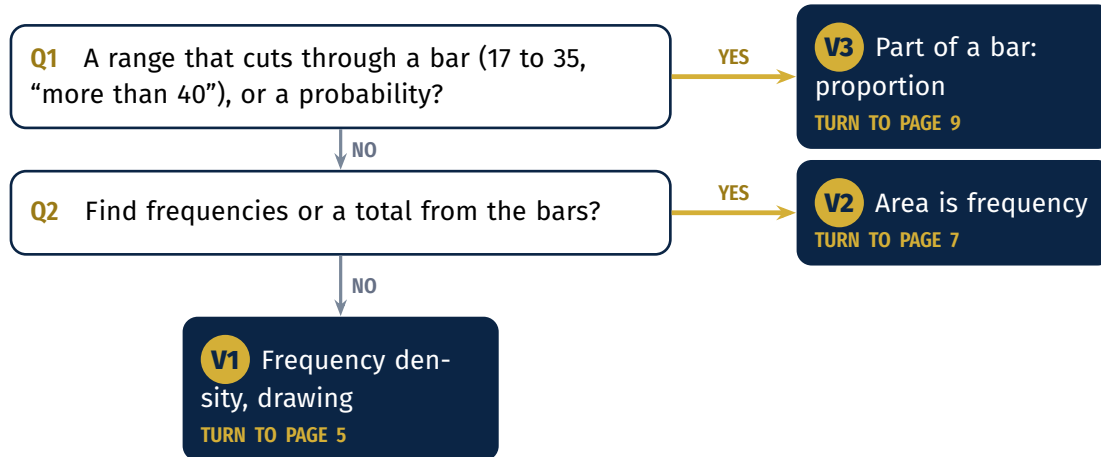
Check: the frequencies from your bars must add to the total.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Frequency Density & Drawing

Height = frequency $\div$ class width; draw bars with no gaps.

SEEN IN PAST PAPERS

6 / 26

PRACTICE QUESTIONS: 15



Your notes

Came up in: Oct/Nov 2020 P1H Q18(a) • May 2021 P1H Q18(b) • Nov 2021 P2H Q14(a) • May 2023 P1H Q18 • May 2024 P1HR Q16 • May 2024 P1H Q22

? What is it?

In plain words

Add a **class width** column and a **frequency density** column. Each bar spans its class exactly and is as tall as its fd.

Completing a histogram: find the scale from a known bar first.

IT LOOKS LIKE

$$0 < h \leq 2$$

$$2 < h \leq 5$$

$$5 < h \leq 10$$

class	f	width	fd
$0 < h \leq 2$	5	2	2.5
$2 < h \leq 5$	12	3	4
$5 < h \leq 10$	18	5	3.6
$10 < h \leq 20$	14	10	1.4
$20 < h \leq 35$	9	15	0.6

How do I do it?

1 Widths

Upper – lower boundary.

2 fd

Frequency $\div$ width.

3 Scale

Label the fd axis (or use the known bar).

4 Draw

Bars from boundary to boundary, no gaps.

fd = $f \div \text{WIDTH}$ Bars touch; ends on the boundaries.

V1 Frequency Density & Drawing

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“On the grid, draw a histogram for this information”

→ **fd column; bars of height fd**

“Complete the histogram”

→ **Scale from a full bar; fd for the missing class**

“Complete the table” from bars

→ **$f = \text{fd} \times \text{width (v2)}$** **WORKED EXAMPLE** **PAST PAPER**

May 2024 • Paper 1HR • Q16 • 3 marks

Tree heights: $0 < h \leq 2$: 5, $2 < h \leq 5$: 12, $5 < h \leq 10$: 18, $10 < h \leq 20$: 14, $20 < h \leq 35$: 9. Draw a histogram.

STEP 1 widths 2, 3, 5, 10, 15 ◀ class widths

STEP 2 $\frac{5}{2} = 2.5$, $\frac{12}{3} = 4$, $\frac{18}{5} = 3.6$ ◀ fd

STEP 3 $\frac{14}{10} = 1.4$, $\frac{9}{15} = 0.6$ ◀ fd

STEP 4 Bars 0–2 at 2.5, 2–5 at 4, 5–10 at 3.6, 10–20 at 1.4, 20–35 at 0.6 ◀ draw

ANSWER the histogram on the “big idea” page

✓ areas

$$5 + 12 + 18 + 14 + 9 = 58 \checkmark$$

WATCH OUT

✗ bar heights 5, 12, 18, 14, 9

✓ heights = fd

Heights equal to frequency make wide classes too big.

EXAM TIP

Write the width and fd columns next to the table before drawing.

Check yourself

calculator first • answers on page 11

1.1 Nov 2021 P2H

Minutes late: 0–10: 8,
10–15: 13, 15–25: 19,
25–40: 24, 40–60: 6.
Frequency densities?

Answer:

1.2 Oct/Nov 2020 P1H

Plant heights: 10–20:
35, 20–35: 45, 35–50:
75, 50–70: 40, 70–80: 8.
Frequency densities?

Answer:

1.3 Dr Eslam

$0 < t \leq 10$: 12,
 $10 < t \leq 15$: 20,
 $15 < t \leq 35$: 16.
Frequency densities?

Answer:

V2

VARIANT 2

Area is Frequency

Frequency = $fd \times \text{width}$; with no scale, use a known bar to find what one square is worth.

SEEN IN PAST PAPERS

9 / 26

PRACTICE QUESTIONS: 19



Your notes

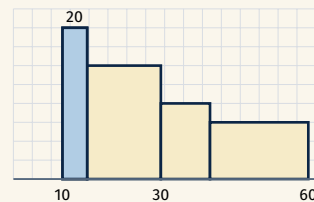
Came up in: Jan 2020 P1HR Q19 • Jan 2021 P1HR Q18 • Jan 2021 P2H Q15(a) • May 2021 P1H Q18(a) • Jan 2022 P1HR Q21 • Jun 2022 P1HR Q19 • Jun 2022 P1H Q21 • Nov 2023 P1H Q17 • Nov 2025 P2H Q21

? What is it?

In plain words

Each bar's **area** is its frequency: $fd \times \text{width}$.

Often there are **no numbers** on the fd axis. One bar's frequency is given: count its **squares** and work out what **one square** is worth. Then count squares in the bars you need.



IT LOOKS LIKE

20 trains in 10–15

How do I do it?

1 Scale

Labelled? Read fd . Not? Use the known bar.

2 One square

Known frequency $\div$ its squares.

3 Each bar

Width $\times$ fd (or squares $\times$ value).

4 Add

Totals, or use for a mean.

AREA = FREQUENCY

One known bar sets the scale.

V2 Area is Frequency

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“There are 160 students with a height between 155 and 170 cm”

→ That area = 160: find the scale

“Work out the total number ...”

→ Add every bar's area

 WORKED EXAMPLE DR ESLAM

Area is frequency • 3 marks

No scale on the fd axis. 20 trains were 10–15 min late; that bar is 8 squares tall. The 15–30 bar is 6 squares, 30–40 is 4, 40–60 is 3. How many were at least 25 min late?

STEP 1 $fd = \frac{20}{5} = 4 = 8 \text{ squares} \Rightarrow 1 \text{ square} = 0.5$ ◀ scale from the known bar

STEP 2+3 $5 \times 3 + 10 \times 2 + 20 \times 1.5$ ◀ 25–30 (part), 30–40, 40–60

STEP 4 $= 15 + 20 + 30 = 65$ ◀ add

ANSWER 65 trains

✓ all areas use the same scale ✓

 WATCH OUT

✗ counting heights only

✓ width × fd

A wide short bar can hold more than a tall thin one.

 EXAM TIP

Write each bar's frequency on top of it before answering.

 Check yourself

calculator first • answers on page 11

2.1

Dr Eslam

A bar from 20 to 50 has frequency density 1.4. How many values in this class?

Answer:

2.2

Dr Eslam

The 10–15 bar is 8 squares tall and holds 20. The 15–30 bar is 4 squares tall. How many in it?

Answer:

2.3

Practice • Hard

74 people: classes give 10, 30, 20; the 170–180 class holds the rest. Height of its bar?

Answer:

V3**VARIANT 3** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

11 / 26

PRACTICE QUESTIONS: 6

Part of a Bar: Proportion & Probability

Split a bar in proportion to its width; add the areas wanted; divide by the total.

Came up in: Jan 2020 P2H Q18 • Oct/Nov 2020 P1H Q18(b) • Jan 2021 P2H Q15(b) • May 2021 P1H Q18(c) • Nov 2021 P2H Q14(b) • Jan 2022 P2H Q20 • May 2023 P1HR Q22 • Nov 2024 P1H Q18 • Jun 2025 P1H Q17 • Jun 2025 P2HR Q20


Your notes

? What is it?

In plain words

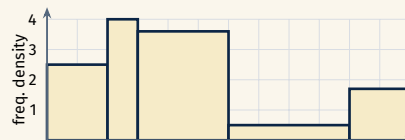
“More than 40” may cut a bar in two.
Assume the values are
spread evenly: the part from 40 to 50 has area $fd \times 10$.

Proportion or probability
$$= \frac{\text{area wanted}}{\text{total area}}$$

IT LOOKS LIKE

more than 40 min

between 20 and 60



How do I do it?

1 Frequencies

Every bar: $fd \times \text{width}$; total.

2 Cut

Part of a bar: $fd \times \text{the part's width}$.

3 Wanted

Add the areas in the range.

4 Divide

By the total (or the given group for “given that”).

PART WIDTH $\times$ fd The most common variant here.

V3 Part of a Bar: Proportion & Probability

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"proportion ... in the shop for more than 40 minutes" → **Cut the 30–50 bar at 40**

"Given that ... more than 30 minutes" → **Divide by the frequency above 30 only**

"Give your answer as a percentage" → **×100**

WORKED EXAMPLE **PAST PAPER**

June 2025 • Paper 1H • Q17 • 3 marks

Histogram fd: 0–10: 2.5, 10–15: 4, 15–30: 3.6, 30–50: 0.5, 50–60: 1.7. Estimate the proportion of people in the shop for more than 40 minutes.

STEP 1 25, 20, 54, 10, 17; total 126 ◀ all frequencies

STEP 2 40–50: $10 \times 0.5 = 5$ ◀ part of a bar

STEP 3 $5 + 17 = 22$ ◀ more than 40

STEP 4 $\frac{22}{126} = \frac{11}{63}$ ◀ proportion

ANSWER $\frac{11}{63}$

✓ about 17% ✓

WATCH OUT

✗ using all of the 30–50 bar

✓ only 40–50: half of it

Cut the bar at 40.

EXAM TIP

"Given that" changes the denominator — read it twice.

Check yourself

calculator first • answers on page 11

3.1 Nov 2024 P1H

fd: 0–10: 1.5, 10–25: 0.8, 25–30: 4, 30–60: 2.5, 60–65: 4.6. Fraction taking between 20 and 60 minutes?

Answer:

3.2 Nov 2021 P2H

Late flights: 0–10: 8, 10–15: 13, 15–25: 19. A flight late by 25 min or less is chosen. P(late by 5 min or less)?

Answer:

3.3 Dr Eslam

The class $20 < x \leq 40$ has frequency 30. Estimate how many are between 20 and 25.

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked in Python (check_answers.py).

1.1

0.8, 2.6, 1.9, 1.6, 0.3

1.2

3.5, 3, 5, 2, 0.8

1.3

1.2, 4, 0.8

2.1

42

2.2

30

2.3

1.4

3.1 $\frac{99}{145}$ **3.2** $\frac{1}{10}$ **3.3**

about 8

THE WHOLE TOPIC ON ONE PAGE

Histograms cheat sheet



Your notes
V1 Draw 6/26

$$fd = \frac{f}{w}$$

Bars touch, on the boundaries.

V2 Area 9/26

$$f = fd \times w$$

Known bar sets the scale.

V3 Part of a bar 11/26

$$\frac{22}{126}$$

Part width $\times$ fd; $\div$ total.**TOP 3 LOST MARKS**

1. height used as frequency $\times$
2. the whole bar when only part is in range $\times$
3. wrong total for "given that" $\times$

BOTTOM LINE Area = frequency $\cdot$ fd = $f \div$ width $\cdot$ one bar sets the scale $\cdot$ cut bars by width.